

MIRI, Malaysia

WMO: 964490

Lat: 4.322N

Lon: 113.987E

Elev: 18

StdP: 101.11

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	22.9	20.8	15.5	26.3	21.3	16.0	25.9	7.3	28.9	6.5	28.9	1.7	100	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.9	32.9	26.6	32.2	26.5	32.0	26.5	27.9	31.2	27.6	30.8	27.2	30.5	3.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.8	30.1	26.6	22.2	29.6	26.2	21.7	29.1	89.8	31.2	87.5	31.2	86.3	31.0	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.3	4.8	DB	20.9	34.2	1.8	0.6	19.6	34.7	18.5	35.0	17.5	35.4	16.2	35.8	(4)
			WB	20.6	29.1	1.0	1.0	19.9	29.8	19.4	30.4	18.8	30.9	18.1	31.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.9	27.0	27.4	27.8	28.0	27.8	27.6	27.7	27.6	27.3	27.3	27.2
	DBStd	0.94	0.99	0.95	0.88	0.82	0.91	0.88	0.90	0.94	0.85	0.85	0.74	0.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6373	523	476	539	534	558	533	546	547	527	537	519	533
	CDD18.3	3331	265	242	281	284	300	283	287	289	277	279	269	274
	CDH23.3	33509	2316	2145	2735	3027	3318	3016	3004	3040	2889	2797	2640	2582
	CDH26.7	11253	631	576	859	1071	1263	1108	1105	1111	1018	917	828	766
Wind	WSAvg	2.0	2.1	2.2	2.0	1.9	1.9	1.9	2.0	2.1	2.1	2.1	2.0	2.0
Precipitation	PrecAvg	2845	324	160	157	173	194	212	202	220	223	302	316	353
	PrecMax	3453	1734	379	365	398	377	649	563	370	429	499	703	694
	PrecMin	2071	7	2	7	5	27	62	37	42	59	118	53	126
	PrecStd	370	284	108	90	79	94	107	102	92	97	101	115	138
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.1	32.6	32.9	33.3	33.1	33.1	33.1	32.9	32.3	32.2	32.2
		MCWB	26.5	26.1	26.3	27.1	27.0	26.4	26.2	26.1	26.6	26.2	26.6	26.5
	2%	DB	31.1	31.1	31.8	32.1	32.6	32.2	32.2	32.2	32.1	31.8	31.2	31.3
		MCWB	26.2	26.1	26.3	26.8	27.0	26.4	26.3	26.3	26.3	26.3	26.4	26.4
	5%	DB	30.3	30.3	31.0	31.6	32.0	31.9	31.8	31.8	31.5	31.1	30.9	30.9
		MCWB	25.9	25.9	26.3	26.6	26.8	26.3	26.3	26.3	26.2	26.2	26.3	26.3
	10%	DB	29.8	29.8	30.3	31.0	31.4	31.2	31.1	31.1	30.9	30.5	30.2	30.1
		MCWB	25.7	25.7	26.0	26.5	26.7	26.3	26.1	26.1	26.1	26.0	26.0	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.2	27.6	28.1	28.7	27.9	27.9	27.7	28.0	27.6	27.6	27.5
		MCDB	30.4	30.3	30.7	31.4	31.5	31.4	31.5	31.2	31.2	30.7	30.7	30.6
	2%	WB	26.7	26.7	27.1	27.6	27.9	27.3	27.2	27.2	27.2	27.0	27.0	27.0
		MCDB	29.8	29.7	30.3	30.8	31.0	30.9	30.9	30.7	30.6	30.4	30.1	30.2
	5%	WB	26.3	26.3	26.7	27.1	27.4	26.9	26.8	26.8	26.7	26.6	26.6	26.6
		MCDB	29.2	29.1	29.7	30.3	30.6	30.5	30.4	30.2	30.1	29.8	29.6	29.6
	10%	WB	26.0	26.0	26.3	26.7	27.0	26.5	26.4	26.4	26.4	26.2	26.3	26.2
		MCDB	28.7	28.8	29.1	29.7	30.3	29.9	29.9	29.7	29.6	29.3	29.1	29.0
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.1	6.5	6.6	6.9	7.0	7.1	7.0	6.9	6.6	6.5	6.3
		MCDBR	6.8	6.8	7.2	7.3	7.6	7.8	7.8	7.8	7.6	7.3	7.1	7.1
	5% WB	MCWBR	2.8	2.7	2.9	3.1	3.2	3.2	3.3	3.3	3.3	3.1	3.1	2.9
		MCWBR	6.4	6.2	6.6	6.8	7.1	7.2	7.2	7.2	7.1	6.8	6.6	6.6
Clear-Sky Solar Irradiance	taub	0.423	0.435	0.434	0.423	0.408	0.414	0.415	0.460	0.458	0.465	0.415	0.418	
	taud	2.471	2.416	2.416	2.446	2.501	2.489	2.483	2.331	2.333	2.321	2.481	2.490	
	Ebn at Noon	900	894	892	885	878	861	865	841	857	857	901	898	
	Edh at Noon	114	122	123	117	107	107	108	129	131	133	112	110	
All-Sky Solar Radiation	RadAvg	4.71	5.27	5.67	5.68	5.31	5.04	5.09	5.18	5.09	4.79	4.82	4.67	
	RadStd	0.44	0.37	0.27	0.22	0.21	0.21	0.25	0.17	0.31	0.29	0.26	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+0.27	+0.23	+0.20	N/A	N/A	N/A	N/A

Nomenclature: See separate page